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Subject : DSA

Unit 3, S12, SLO1 - Circular Queue - Dequeue Operation

Algorithm to delete an element from a circular queue

Step 1: IF FRONT=_____

Write UNDERFLOW

Goto Step 4

Step 2: SET VAL = _____

Step 3: IF FRONT = REAR

SET FRONT = REAR=_____

ELSE

IF FRONT = _____

SET FRONT = 0

ELSE

SET FRONT = _____

Step 4: EXIT

Answer: Step 1: IF FRONT = -1

Step 2: SET VAL = QUEUE[FRONT]

Step 3: IF FRONT = REAR

SET FRONT = REAR = -1

ELSE

IF FRONT = MAX - 1

SET FRONT = 0

ELSE

SET FRONT = FRONT + 1

Step 4: EXIT

Fill in the Missing Programming Code for Circular Queue delete_element():

Answer: int delete_element() {

int val;

if (front == -1) {

printf("\n UNDERFLOW");

return -1;

}

val = queue[front];

if (front == rear)

front = rear = -1;

else {

if (front == MAX - 1)

front = 0;

else

front = front + 1;

}

return val;

}